

General Industries Hub

Enterprise Inventory Management

Warehouse Portal Manual

Receive goods, prepare delivery notes, returns to vendor.

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GI Hub - Role booklet - Warehouse Portal Manual

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no running total can drift out of step with reality.
Two-stage approval	Store Keepers stage entries -> HOD reviews and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal request + approval.
Audit-first	Every consequential action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper confirms it arrived. Every handover is recorded.
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Principle	What it means
Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.

1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock record	Permanent - stock figures move now
4	System	Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then	Warehouse - Receiving

Step	Role	Action	Lands in
		records the physical arrival from the vendor	
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on site	Permanent stock record; the delivery closes

Three side-paths exist for when things do not go to plan:

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Live | Practice switch - above the username. Live is the real system and is selected unless you choose otherwise. Practice is a separate training copy with made-up data, where nothing you do affects Live. Switching signs you out. See §26 Practice Mode for everything about it.
- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open. This works from any address, including a bookmarked /login; an address that is not a page takes you to your home page instead of a blank screen.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.1.1 The look of the sign-in page

Phase 14e, redesigned in Phase 15c.

The large gold GI mark sits in its own space at the top of the page, and the glass sign-in card sits below it, so the two never overlap. The mark rises into place once when the page opens. After that it stands still, and a slow band of light sweeps across the gold every few seconds. Nothing on the page follows the mouse. What you see depends on your device, and the form works the same on all of them, from the first moment:

- Everyone, including the Android, iOS and desktop apps: the mark drawn in the page's style.
- A desktop computer with a graphics card: the same mark in 3D gold, in the same place. It loads separately after the page is ready, so it never delays signing in. It draws nothing between sweeps, stops when you switch tabs, and is removed as soon as you sign in.
- A very short window (under about 560 pixels tall): the large mark gives way to a small, still mark inside the card.
- Anyone with Reduce motion switched on in their system settings: no rise and no sweep. The mark simply stands there.

The Executive Summary and the Training page use the same glass style. Their cards rise gently into place as you scroll, except with Reduce motion on and when printing.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below minimum (or "All levels adequate" if clear)
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
Standard header	Live Dashboard, Entry Log	Gold subtitle accent + current date
HOD header	HOD Portal	Purple subtitle accent + current date
Admin header	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

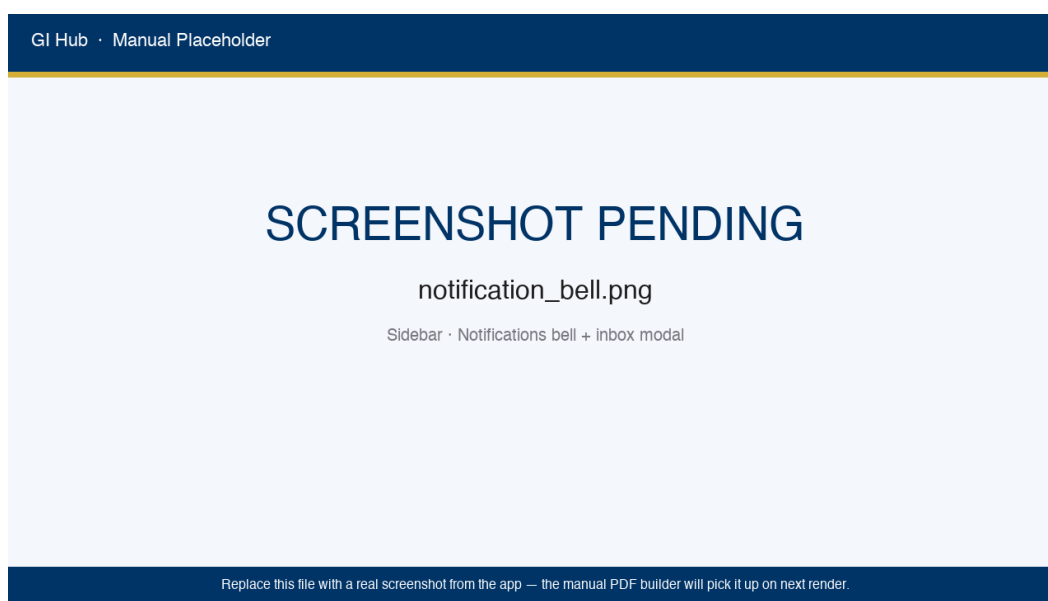
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

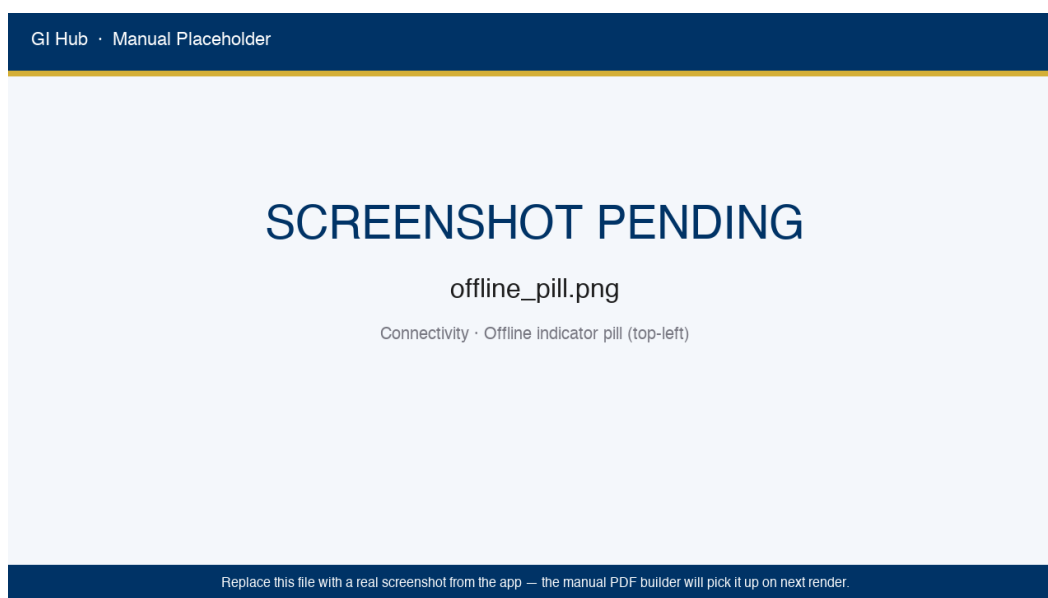
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line

Event you'll see	Triggered by
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if the notification lookup fails, the badge silently shows 0 instead of breaking the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by the per-event WhatsApp toggles in the system settings (see §9).

3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

3.7 Saving an entry when the signal drops

If the network drops while you save a receipt, issue, return, stock adjustment or bulk batch, GI Hub keeps the entry on your device and shows a saved offline message instead of an error. A badge in the header shows how many entries are waiting. They are sent by themselves when the connection returns, and you can press the badge to send them straight away.

An entry is never saved twice. Each save carries its own reference, and the server recognises it if it arrives again. That can happen when the page reloads while the entry is being sent, or when the connection drops just as the server confirms. The second arrival is matched to the first, and only one entry reaches your HOD. You never need to check for, or delete, a duplicate caused by a bad connection.

3.8 What's new

Phase 14d.

When a feature you use changes, a What's new panel opens by itself the next time you sign in. It shows only what concerns you: an announcement is sent to the roles that can open the page it is about, so nobody is told about a screen they cannot reach.

Each card says what changed. Open it takes you to the page, Watch the tutorial plays the short video when there is one, and a User Manual § tag names the section to read.

Press Got it and the panel closes. It does not open by itself again for the same announcements. To see recent ones, press the gift icon in the top bar, next to the bell; a number on it means something new is waiting.

A matching notification also appears in the bell (§3.6). Announcements are in-app only - never WhatsApp or email.

3.9 Stock vs the Excel workbook - which materials differ, and why

Added 26 September 2026.

GI Hub keeps its own stock ledger, and the site also keeps CNCEC_Inventory.xlsx. After every Excel sync, GI Hub compares its stock for every material with the workbook's Current Stock column. It then works out why any material differs.

Where: the top of the Stock page.

- A yellow banner says how many materials differ, for example 14 of 505 materials don't match the Excel workbook, and when that was checked.
- Each of those materials has a red row and a red != Excel tag. Point at the tag to see the reason.
- Review the differences lists every material. Each one shows its workbook and GI Hub figures, what caused the difference, where to look (a GI Hub record number or an Excel sheet and row) and how to fix it.
- A material's own page (scan its QR or open it from Stock) shows the same warning.

What the causes mean:

Tag	What happened	How to fix it
Only in GI Hub	A receipt, issue or return was entered in GI Hub, and the workbook's log never got it. The entry names who made it and its remarks.	If it really happened, add it to the log. If it was a test or a mistake, reverse it in GI Hub with a Stock Adjustment, or have an admin remove it.
No longer in Excel	GI Hub took a line from the workbook, and the workbook has since lost it (deleted or re-dated).	Put the line back if that was a mistake. Otherwise an admin removes it with the sync's --prune-vanished option.
Not synced yet	The workbook has a line GI Hub doesn't have yet.	Run the Excel sync.
Edited in Excel	A quantity was changed in the workbook after the last sync.	Run the Excel sync.
Sheet totals stale	The Inventory sheet's own Current Stock doesn't match its own logs. Current Stock is a typed value in this workbook, so it can fall behind.	Correct the figure on the Inventory sheet.
Opening stock	The Opening Stock differs.	Run the sync, or correct the workbook.
Unexplained	Part of the difference that none of the above explains. It is always shown, never hidden.	Compare the material's records by hand.

The marked workbook. Press Get the marked workbook and choose your CNCEC_Inventory.xlsx. You get back a copy containing:

- a GI Hub check sheet first, listing every cause with how to fix it;
- each differing material's row on the Inventory sheet filled red, with a note on its Current Stock cell;
- any log line the check names filled amber.

Make the fixes in your own workbook, then sync again. Your file itself is never changed. Check again uploads the workbook and refreshes the list without downloading anything.

Store Keepers, Warehouse, HODs, Logistics and Admins can upload. Everyone who can see Stock can read the result. Nothing on this screen changes stock.

[!] The check only explains. It never corrects stock by itself, because deciding whether an entry was real is a person's call.

3.10 Lots, FEFO and the Lot Register workbook

Phase 16 (October 2026).

A lot is one batch of a material: the cans of PU COMP A from batch 3504, or the CHEMOLINE rolls of batch 1O25003382. GI Hub keeps the stock of each lot, so it can tell the store keeper which lot to use first (FEFO, first expire, first out).

3.10.1 What the `Serial No.` column means in the workbooks

The Receipt, Consumption and Return Logs have one Serial No. column. GI Hub reads it according to the item on that row:

The item is...	Serial No. is...	GI Hub keeps it as...
a Surface Shield (PU, Furan, Phenacin, Coroflake, BC 3004...)	the batch	the Lot No. (and the cell as typed)
a roll (CHEMOLINE, UOM ROL)	the roll number	the roll number, and its batch as the Lot No.
a brick (AR 30/40 MM, carbon bricks)	not a lot - bricks have no batch and never expire	the serial, with no lot
anything else (pumps, tools, equipment)	the equipment / asset no.	the serial, as before

Rules that keep this safe:

- A lot is spelled one way. 3504.0 is 3504, and A4525 and A 4525 are the same lot (A 4525). N/A or a blank cell means no lot.
- Roll numbers begin with 1O (the letter O). A roll typed 10... (with a zero) is corrected to 1O..., and the sync says so.
- One lot per cell. A cell such as 4525 = 55 Cans; 1823 = 2 Cans gets no lot. Split it into one row per lot in the workbook; the sync names the rows that need it.
- A lot that was never received is reported, not invented. If the Consumption Log names a lot no receipt brought in for that material, the sync lists it with the material it was received under (often a typing slip), and the consumption still counts toward stock.

- History keeps the lot it was given. GI Hub never picks a lot for an old workbook row that has none. Automatic FEFO applies only to issues entered in the app.
- A return gives back to its lot. The Return Log's Serial No. is now read too: a Surface Shield return carries its lot, and an equipment return keeps its asset tag.

Every receipt that names a lot creates that lot in GI Hub. A lot's stock is always received - consumed - returned, worked out from the ledger.

3.10.2 The Lot Register workbook ('Rubber & Brick Materials - CNCEC.xlsx')

This workbook keeps, per Surface Shield material, one sheet of what arrived: batch, manufacture date (MFD), expiry, DN No. and quantity. Put it in the same folder as the other workbooks. The normal sync reads it automatically; any file named like *Rubber*Brick*CNCEC*.xlsx is found, and the run prints a > lots section.

[!] It never changes stock. Stock comes only from the Receipt, Consumption and Return Logs. This workbook only describes the lots GI Hub already has: their MFD, expiry and batch. Its quantities are compared with the Receipt Log, and every difference is reported. A mistake in it can make an expiry wrong, but it can never double a quantity.

How each row is matched:

1. To its material - by the SAP column. If a sheet has no SAP column, GI Hub matches the material name, the component letter (COMP C) and the pack size (10 Kg vs 2.5 Kg), and refuses anything it cannot pin to exactly one item. The material code is only a cross-check, because COMP A-D share one code.
2. To its receipt - the same SAP, DN No. and received date. CHEMOLINE rolls have no DN, so they match on SAP and date, roll by roll.
3. To its lot - the Batch No. (normalised as in §3.10.1). For CHEMOLINE the Order No. is the batch and each Roll No. is a roll inside it.

What it fills in:

On...	It writes...
the lot	MFD, expiry, batch reference, DN No.
the roll register	each CHEMOLINE roll and the batch it belongs to
a receipt with no lot	the lot, when the workbook names exactly one batch for it
an item with no shelf life	the shelf life (months), learned from its own MFD -> expiry pairs

Expiry:

- An expiry in the workbook is used as it is (from the file).
- A missing expiry is MFD + the item's shelf life (derived).
- If two rows of one lot disagree, the earliest expiry is used.
- An expiry before its MFD is refused and reported.
- An expiry a person typed on the Receive form is never overwritten.

Bricks are compared with the receipts but get no lot (they have no batch and never expire).

What the sync tells you (each is listed by sheet and row):

- quantities that differ from the Receipt Log;
- rows with no matching receipt;
- expiries before the MFD;
- lots whose rows disagree on dates;
- material codes that disagree with the inventory;
- a batch the Receipt Log spells differently (e.g. 0926 vs 0.926);
- rows with no batch.

3.10.3 Lots & Expiry - which lot to use, and what is about to expire

The Lots & Expiry page (sidebar, under Stock) is for the Store Keeper, QC, the HOD and the Head of Qualities. It lists every lot of a lot-tracked material, the oldest expiry first, with:

- MFD and expiry - an expiry marked derived is $MFD + \text{the item's shelf life}$;
- status - Expired, ≤ 30 days, ≤ 60 days, ≤ 90 days, OK, No expiry, or Quarantined / Disposed when the Admin has taken the lot out of use (Admin Console -> Lots). A quarantined or disposed lot is never offered on the Issue form;
- received, consumed, returned and remaining - worked out from the ledger.

Click a status at the top to show only those lots. Show used-up lots adds the lots with nothing left. A CHEMOLINE batch opens to show its rolls: when each was received, where it is, and for a used roll the day and tank it was issued to. A red banner warns when an expired lot still has stock. At the bottom, Lots used but never received lists lots the Consumption or Return Log names that no receipt brought in.

Issuing (the Issue form's Lot field). For a lot-tracked material, the Lot field is a list of that material's open lots in FEFO order. Each shows its expiry, the days left and the quantity left.

- The first one is marked FEFO. Leaving the field blank issues from it, because GI Hub's own automatic pick uses the same order.
- [!] An expired lot is listed last, marked red, and is never the suggestion, even though its date is the earliest. You can still choose it; FEFO warns, it never blocks.
- Choosing any lot other than the FEFO lot asks for a reason, which goes to the HOD with the issue. Choosing the FEFO lot asks nothing.
- CHEMOLINE: pick the roll; its batch fills in by itself. A roll already issued is no longer offered.

Receiving (the Receive form). For a lot-tracked material the form asks for the batch (as printed on the label), the manufacture date (MFD) and the expiry:

- If you leave the expiry blank, it becomes $MFD + \text{the item's shelf life}$, and the form shows the date it will use.
- An expiry before the MFD, or an MFD in the future, is refused.
- An expiry you type is kept, and the Lot Register workbook never overwrites it.

The daily expiry notice. Each evening the Store Keeper and the HOD of a site get one notice listing that site's lots that expire within 30 days or have already expired, each with how much is left. It is one notice per site, never one per lot. On WhatsApp it arrives inside the evening summary.

Shelf life and lot tracking (item editor). Admin -> Inventory or Records -> Inventory -> Edit has two

settings:

- Lot tracking: Automatic - every Surface Shield except bricks - or Tracked by lot, or Not tracked.
- Shelf life (months): the number a missing expiry is derived from. The Lot Register workbook fills it in where it is empty, from the item's own MFD -> expiry pairs (PU 9, Phenacin 6, BC 3004 24...).

Reading the two warnings - they are different things.

You see...	It means...	What to do
a red Expired lot with stock left	The lot was received, its expiry has passed, and the ledger says some of it is still on the shelf.	Check the shelf. If it is there, ask the Admin to dispose of it or quarantine it. If it is not, an issue or return was never entered - enter it, or correct the workbook.
a lot under Lots used but never received	The Consumption or Return Log names a lot that no receipt of that material brought in. The lot itself may be fine; usually the lot number or the SAP code on that line was typed wrong. The column Received under SAP shows the material that did receive a lot with that number.	Correct the line in the workbook (the lot, or the SAP code) and sync again. The stock already counts the line either way.

An expired lot whose stock is all used shows Used up, not Expired, and appears only with Show used-up lots switched on. So when a lot shows received 1, consumed 1, remaining 0, there is nothing to dispose of.

3.10.4 Practising with lots

Practice (\$26) has lots to try all of this on:

- PRACTICE PU PRIMER (899971) has three lots: PR-OLD (expired, still has stock), PR-SOON (expires within 30 days, so it is the FEFO suggestion) and PR-LATE.
- PRACTICE CHEMOLINE (899973) has batch 1O26009999 with three rolls to pick.
- One issue names lot PR-TYPO, which no receipt brought in, so the Lots & Expiry page has a used but never received example.

3.11 What changed in September-October 2026 (Phases 14, 15 and 16)

A one-page list for everyone who used GI Hub before September 2026. Each line names who it affects and where it is explained in full.

3.11.1 Phase 14 - quantities, reconciliation and announcements (26 September)

Change	Who	Read
Surface Shield quantities are shown both ways, kilograms first (189 KG - 21 Can). You still enter packs; a consumption form may be written in KG and is converted.	Store Keeper, Supervisor, HOD	§4.11, §4.10.8
When a QR form and the Excel workbook both record the same drum, the ledger keeps one quantity (the larger), never both.	HOD	§6.17
The Surface Shield queue is grouped: one job = one tank on one day, area asked once and credited once.	Store Keeper, HOD	§4.9a.2
What's new notices after an update, and tutorial freshness warnings for admins.	Everyone; Admin	§3.8, §7.14
The new 3D login (a still crest on phones and older computers).	Everyone	§3.1
Stock vs the Excel workbook: the Stock page marks materials whose stock differs from the workbook's, and says why.	Store Keeper, HOD, Admin	§3.9
An entry saved while the signal dropped is sent once, however many times it is retried.	Store Keeper	§3.7

3.11.2 Phase 15 - Practice, Garnet and the store keeper's note (28-30 September)

Change	Who	Read
Practice is violet with a pulsing badge, and refuses to start on an out-of-date database (no more half-working pages).	Everyone	§26.2, §17.9
Garnet is surface preparation: it credits no area of its own; the HOD answers Old or New surface per job.	Store Keeper, HOD	§4.9a.7
The store keeper's note fills the job card: Floor - 13.37 SQM Done fills the area, the part and the remark. Two notes on one tank and day are two jobs. The remark goes to the	Store Keeper, HOD	§4.9a.2

Change	Who	Read
HOD, the job history and the exports.		
The HOD adds and edits inventory items for their own site (Records -> Inventory -> New item / Edit). Deleting stays Admin-only.	HOD	§2.3.4, §2.3.8-2.3.9 (Adding an inventory item)
A user who belongs to one site no longer sees a Site box on the forms: their site is filled in and shown, not asked. Admins and multi-site users still choose.	Store Keeper, HOD	this section
The WBS No. field appears on Receive and Issue. If the HOD has not set up any WBS numbers yet it shows "None set up" and the entry still saves.	Store Keeper, HOD	§13.14, §16.5
Site and Category on a new item are matched to the spelling already in use (cncec -> CNCEC).	Admin, HOD	§2.3.8-2.3.9 (Adding an inventory item)

3.11.3 Phase 16 - lots, FEFO and the Lot Register workbook (1 October)

Change	Who	Read
The workbooks' Serial No. is read by the item: the batch for a Surface Shield, the roll for CHEMOLINE, the asset tag for equipment. Live now has real lots.	Everyone	§3.10.1
The Lot Register workbook (Rubber & Brick Materials - CNCEC.xlsx) adds each lot's MFD, expiry and rolls. It never changes stock.	Admin	§3.10.2
Lots & Expiry page: every lot, its expiry, and what is left of it.	Store Keeper, QC, HOD, Head of Qualities	§3.10.3
The Issue form's Lot field lists lots in FEFO order; an expired lot is listed last and never suggested. Choosing another lot asks for a reason (it never blocks).	Store Keeper	§3.10.3
The Receive form asks for the MFD; a blank expiry becomes MFD + shelf life.	Store Keeper	§3.10.3

Change	Who	Read
One evening expiry notice per site lists lots expiring within 30 days.	Store Keeper, HOD	§3.10.3
A return gives back to its lot, so a lot's balance is received - consumed - returned.	Everyone	§3.10.1
Practice has example lots: an expired one, a FEFO one, CHEMOLINE rolls, and a lot used but never received.	Everyone (training)	§3.10.4

3.12 The Hub Assistant answers some questions directly (Phase 17, October 2026)

The Hub Assistant (the round button at the bottom right of every page) now takes a quick look at your question before answering it - about half a second - and decides what kind of question it is. Four things can happen:

You ask...	You get
to open a page - "open the lots page", "go to Reports"	An Open ... button for that page. Only pages your role can open are ever offered; a page you cannot open, or a name that fits several pages, gets an ordinary answer instead. It only opens pages - it never does anything for you.
for a video - "is there a video on staging a return?"	The Watch it button, straight to the moment in the training video that shows it. If no video covers it, you get an ordinary written answer.
for numbers or records - "show me the receipts from last week", "which lots expire in the next 30 days?"	A small table in the chat. This is the same data as Reports -> AI -> Ask your data, and only the roles that already have that card get it: HOD, Logistics, Auditor and Admin. An HOD sees their own site only. Everyone else gets the usual written answer from the manual.
anything else - how to do something, what a word means, who can do what	A written answer from your section of the manual, exactly as before.

A question that tries to trick the assistant - telling it to ignore its rules, asking for its hidden instructions, typing database commands - is refused with the usual sentence: "I can only answer questions about your section of the manual." Ordinary work sentences that happen to use the same words ("drop the damaged drums at bay 3", "ignore the damaged drum and issue the rest") are answered normally.

If the quick check is not available (the AI service is restarting, or an admin has switched it off), the assistant simply answers as it always did. You lose nothing; only the buttons and tables do not appear.

For admins: the quick check can be switched off without a restart - Admin -> Console -> Settings, set `ai_router_enabled` to 0 (and back to 1). The AI health line shows whether it is on and ready.

Two smaller changes arrived with it:

Change	Who	Read
The Head of Qualities no longer sees Reports or the Records ledgers (Receipts, Consumption, Returns, Lots, Purchase Requests) in the menu. They never worked for this role - every one showed an error - and the role's pages are Quality Oversight and Lots & Expiry.	Head of Qualities	§2.2, §2.3.5
The Watch it button appears less often by mistake. A question that merely says "video" or "tutorial", or shares only a word like "many" or "into" with a video, no longer gets a link to the wrong moment.	Everyone	§24.2.3

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at site)
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for audit)
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	The Material Code or Description contains "RL-", "RUBBER LINING" or "RUBBER-LINING"
BL	Brick Lining	It contains "BL-", "BRICK LINING", "BRICK-LINING" or "BRICK MATERIAL"
(none)	Neither family	The default when nothing above matches

The system works this out for you. An item is never tagged as both - if the wording matches both families, RL wins.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

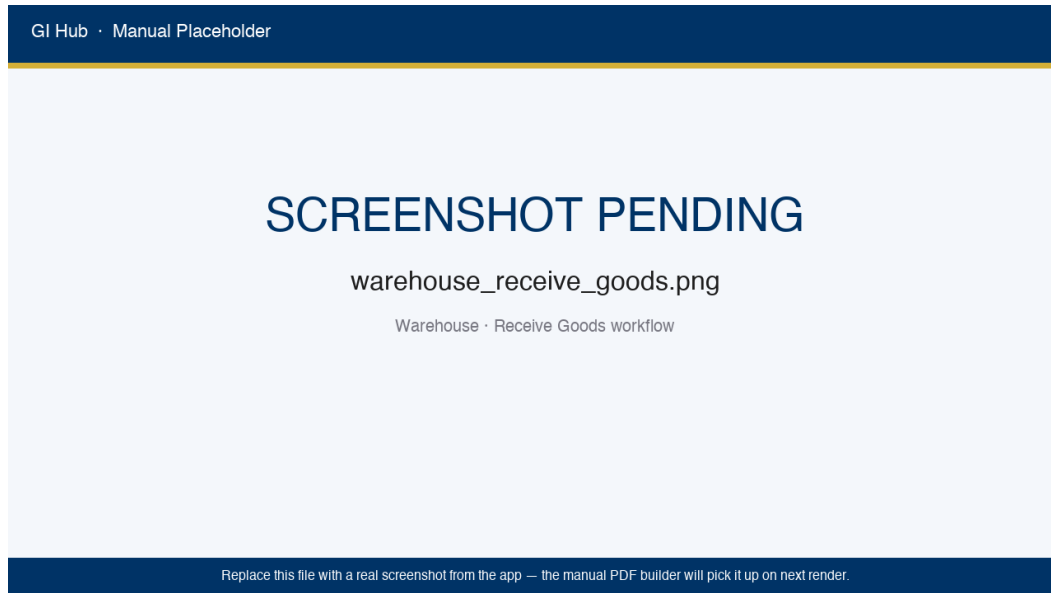
11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's

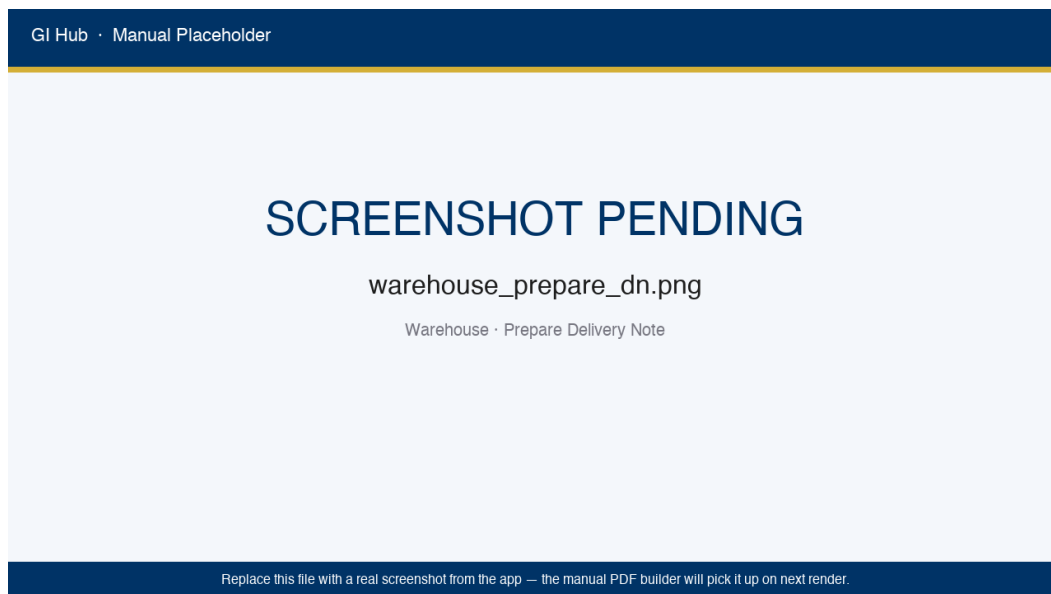
Term	Meaning
	transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

15. Warehouse Portal Manual

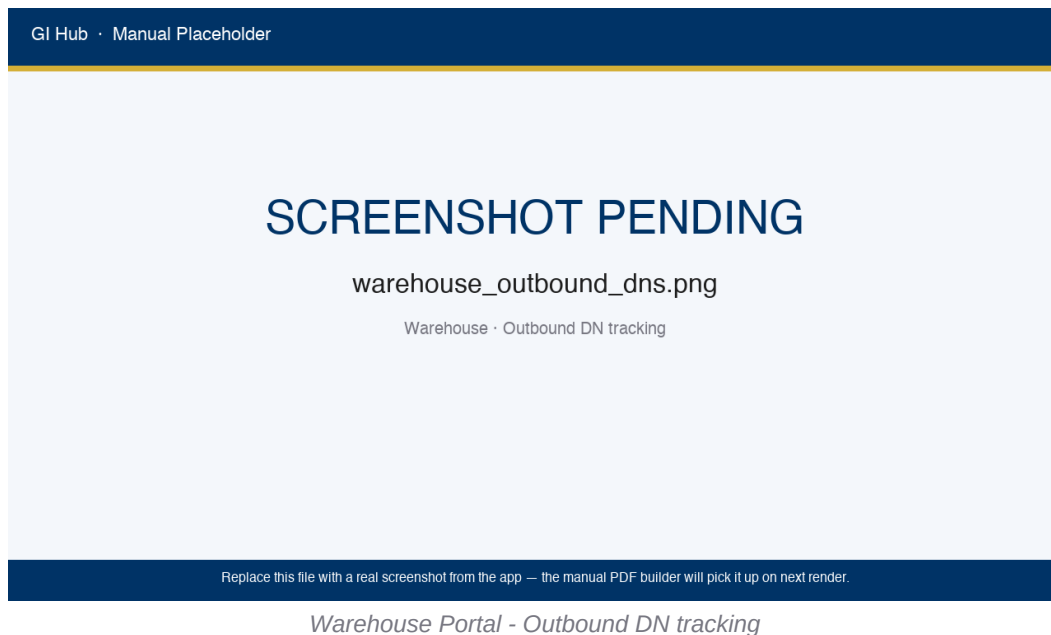
The Warehouse Portal is the physical-receiving and DN-preparation side. Role-locked to {warehouse_user, admin}. Six tabs. Prices are completely hidden in every view - three independent enforcement layers guarantee Unit_Price, Total_Price, Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are never visible to a warehouse user.



Warehouse Portal - Receive Goods workflow



Warehouse Portal - Prepare Delivery Note with RL/BL strict separation



15.1 Pages visible to Warehouse User

- Live Dashboard
- Warehouse Portal (this page)
- Reports
- Locator - which rack a material is on, and the reverse: scan a rack and see everything that should be on it, which turns a stock count into a checklist (see §21.8)
- Assets - serialised tools and equipment, each with its own record, status and location history, including a map link where a position was captured (see §21.7)

15.2 Sidebar warehouse resolution

- A Warehouse User is tied to a single warehouse on their own user record. The portal works out which one from their profile.
- An Admin shadowing the portal gets a " Shadow warehouse" picker in the sidebar listing every active warehouse.
- If neither applies, a red error card appears: " Your account is not bound to a warehouse. Ask Admin to set your Warehouse ID in Admin Portal -> Users."

15.3 Page title

" Warehouse <ID>" in gold, with a muted "(prices hidden - Logistics-only)" caption next to it as a permanent visual reminder.

15.4 Tab 1: Incoming Assignments

POs Logistics has routed to this warehouse.

15.4.1 Hero strip

Card	Source
AWAITING ACK	Assignments routed to you that you have not acknowledged yet
ACKED	Acknowledged - you've seen them, waiting on the goods
IN/RECEIVED	Partly or fully received

15.4.2 Assignments grid

A table with columns: Assign # - PO No. - PR No. - Vendor - Dest Site - Expected - Assigned - Acked - Status. No price columns.

15.4.3 Acknowledge action

A picker listing every assignment still awaiting acknowledgement. Choose one, click [OK] Acknowledge, and it moves to acknowledged and pings Logistics with a "warehouse acknowledged" notification (this one is off by default in the WhatsApp toggles to keep the noise down).

15.5 Tab 2: Receive Goods

Record qty actually received at this warehouse against an acknowledged assignment.

15.5.1 Assignment picker

A picker listing every assignment you have acknowledged but not yet fully received.

15.5.2 PO snapshot card (no prices)

KV row	Value
PO Number	the PO
Vendor	<code> - <name>
Inco / Payment	the terms
Expected	the date

Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are stripped from the header dict before render. Never visible.

15.5.3 Receive grid

Editable grid showing every line on the assignment:

Column	Source
id	po_items.id (disabled - read-only)
Material_Code	line code

Column	Source
Description	line description
UOM	unit
rl_bl_family	family chip (RL / BL / blank)
Qty	ordered (read-only)
Delivered_Qty	already received cumulative (read-only)
Open_Qty	computed Qty - Delivered + Returned (read-only)
Receive Now	EDITABLE - type the qty you physically received this event

15.5.4 Record receipt button

- Validates: at least one row has a Receive Now quantity above zero
- Over-deliver guard: what has been delivered, less anything returned, plus what you are recording now, may NOT exceed the quantity ordered. If it does, the whole batch is rejected with a friendly message naming the line.
- On success: each line's delivered quantity goes up, and the line is marked delivered once the net delivered amount covers the ordered quantity, or partly delivered otherwise
- The assignment moves to received once every line on the parent PO is delivered, and to partly received otherwise
- The PO itself is updated to match
- A "warehouse received goods" notification goes to Logistics with the line count

15.6 Tab 3: Prepare DN

Build a Delivery Note for a site. RL/BL strict separation is enforced here - a DN cannot span both families. If you try, the action is rejected with: "Strict separation violated: this DN spans multiple RL/BL families. Prepare one DN per family."

15.6.1 Source pickers

- PO No. picker - assignments you have acknowledged, partly received or fully received
- Destination Site selectbox

15.6.2 Items grid

Editable grid with read-only inventory columns (Material_Code, Description, UOM, rl_bl_family, Qty, Delivered_Qty, Returned_Qty) and editable shipping columns:

Column	Purpose
Ship Qty	What you're sending on this DN (must be > 0 to include)
Lot_Number	If you tracked the lot at receive time
Expiry_Date	Lot expiry

Column	Purpose
Remarks	Free text

15.6.3 DN header

Three-column row: DN Date - Vehicle No - Driver Name - Driver Phone - Prepared By (auto-filled with your username) - Remarks.

15.6.4 Save DN draft button

- Validates: at least one row with Ship Qty > 0
- Over-ship guard: for each line, what you may ship is what was delivered to you, less anything already returned, less anything already committed on other live Delivery Notes. Ship quantity cannot exceed that.
- RL/BL strict-separation check: if the items span both families, REJECTED
- On success: a new Delivery Note is created as a draft, with one line per item shipped. The DN header records the RL/BL family where the items carry one.
- Toast: DN <number>

15.6.5 DN numbering convention

DN-<WAREHOUSE_ID>-<YYYYMMDD>-<seq> - seq resets per (warehouse, day). Example: DN-WH-A-20260616-003.

15.7 Tab 4: Outbound DNs

Track every DN this warehouse has prepared.

15.7.1 Status filter

Multi-select of every DN status. Defaults to the active ones - draft, awaiting Logistics, awaiting HOD and awaiting Store Keeper.

15.7.2 DN grid

A table of every matching DN with full state metadata.

15.7.3 Submit-to-Logistics action

Whenever the filter shows a draft, a "Submit a draft to Logistics" section appears with a picker of your drafts and a Submit to Logistics button. Submitting moves the DN into the Logistics approval queue and sends them the DN-approval notification.

15.7.4 Per-DN drilldown

Select a DN -> section card showing PO, Site, Status, RL/BL family, Vehicle/Driver, and the full signature trail (Logistics decided by/decision, HOD decided by, SK received by). Items grid below.

15.7.5 Request reschedule expander

If the date you targeted isn't going to work, raise a reschedule from here. Same flow as the HOD's In-Transit tab - you cannot pick a date in the past, and a reason is mandatory.

15.8 Tab 5: <- Returns from Site

Raise a return when a site flags defective material from a DN this warehouse delivered.

15.8.1 DN picker

Selectbox of DNs in received / pending_sk / hod_approved states.

15.8.2 Items grid

Editable grid with a Return Qty column. Filling > 0 on a line includes it in the return.

15.8.3 Reason textarea

Mandatory.

15.8.4 <- Raise return to vendor button

- One return is raised per affected line
- For each line: a return record is created showing it was raised by the Warehouse, the DN line is flagged as returned, the parent PO line's returned quantity goes up, and that line reopens as either partly delivered or open
- If the PO had already been closed, it reopens
- Notification fires to Logistics

15.9 Tab 6: History

Read-only. Two sub-tabs: Completed DNs (status in received / rejected / cancelled) and Closed assignments.

15.10 Warehouse - Use Cases

Use Case 1: Receive a vendor delivery

1. Incoming Assignments -> [OK] Acknowledge the assignment when you see it
2. Goods physically arrive
3. Receive Goods -> pick the assignment -> type the actual received qty per line -> Record receipt

Use Case 2: Ship to a site (single family)

1. Prepare DN -> pick PO + destination site

2. Type Ship Qty per line - keep all RL or all BL
3. Fill DN header (Vehicle, Driver, Date)
4. Save DN draft -> status = draft
5. Outbound DNs -> select the draft -> Submit to Logistics

Use Case 3: Ship RL + BL to the same site on the same day

You need TWO DNs:

1. Prepare DN with only RL lines -> save + submit
2. Prepare second DN with only BL lines -> save + submit

The system rejects any attempt to combine them on a single DN.

Use Case 4: A site rejects part of a DN - raise to vendor

1. <- Returns from Site -> pick the DN
2. Type Return Qty on the offending line
3. Reason: "Site reported defective surface coating"
4. <- Raise return to vendor - PO + po_item reopen; Logistics sees the return in their tab

Use Case 5: Request a reschedule

1. Outbound DNs -> drill into the affected DN
2. Request reschedule for this DN -> pick new date + reason -> Submit

15.11 Warehouse - FAQ

Q: Why don't I see prices anywhere? A: Warehouse role is intentionally blind to commercial data. Three independent enforcement layers strip every monetary field. If you genuinely need a price (e.g. to file a damage claim), ask Logistics or Admin.

Q: My over-ship guard rejected a DN. I just received everything from the vendor. A: Probably another DN is already drafted (or submitted) shipping a slice of that line. Check Outbound DNs filter for live DNs against the same PO. If one was abandoned, ask Logistics to reject/cancel it so the qty frees back up.

Q: RL and BL are going to the same site on the same truck. Can I combine? A: No. Two DNs. The strict separation is by design (different testing standards, different storage requirements). The site receives them as two separate DNs in their Receipt Staging queue.

Q: Where did the "Receive Goods" assignment go? It was here this morning. A: Once every line on the parent PO is fully delivered, the assignment moves from partial to received and stops appearing in Tab 2 (which filters to acknowledged + partial). Check the assignment grid in Tab 1.

Q: Can I edit a DN after submitting to Logistics? A: Not directly. Ask Logistics to reject it (Outbound DNs shows the rejection if it happens) - then it returns to draft for you to edit.

16. Cross-Role Procurement Walk-through

A single happy-path narrative threading all five roles together, useful for onboarding.

16.1 The scenario

Site GI-PS01 needs 50 RL panels and 20 BL bricks. Total 3 working days from PR raise to physical receipt.

16.2 Hour by hour

When	Who	What
Day 1 - 8:00 AM	Site HOD	Opens HOD Portal -> Purchase Requests. Creates 2 PR lines (50 RL panels, 20 BL bricks) with PR Number 3000099999, WBS Number 4003951, Network 4003951-PROJ-A, Plant GI-PS01, Delivery_Date Day 3.
Day 1 - 8:15 AM	Site HOD	Opens the Submit PR(s) to Logistics Portal expander -> multi-selects PR 3000099999 -> Submit Selected to Logistics.
Day 1 - 8:15 AM	(auto)	A "PR submitted to Logistics" notification lands in the Logistics inbox with a red badge, plus a WhatsApp ping if that event is enabled.
Day 1 - 9:00 AM	Logistics	Sees PR 3000099999 in Incoming PRs queue. Drills in - 2 lines, RL + BL. Clicks Use this PR to create a PO.
Day 1 - 9:30 AM	Logistics	Switches to Create PO -> manual entry. Picks vendor "Carborundum Universal" (0000110341), Inco/Payment auto-fill, fills PO Number 4720033030, Unit_Price per line, Total_Price computed. Save PO.
Day 1 - 9:30 AM	(auto)	The PR lines are marked as covered by a PO and disappear from the Logistics queue. The site HOD gets a "PO issued" notification with the

When	Who	What
		vendor and PO number.
Day 1 - 10:00 AM	Logistics	Assign to Warehouse -> picks PO 4720033030 + Warehouse WH-A + Expected Delivery Day 2 evening. Assign.
Day 1 - 10:00 AM	(auto)	A "PO assigned to warehouse" notification lands in the WH-A users' inbox, plus a WhatsApp ping if that event is enabled.
Day 1 - 11:00 AM	Warehouse user (WH-A)	Incoming Assignments -> [OK] Acknowledge assignment #N. Goes about their day.
Day 2 - 5:00 PM	Warehouse user	Physical truck arrives from vendor. Receive Goods -> assignment #N -> type 50 in Receive Now for the RL line, 20 for the BL line. Record receipt. Both lines are marked delivered.
Day 2 - 5:30 PM	Warehouse user	Prepare DN -> pick PO 4720033030, destination GI-PS01. Types Ship Qty 50 for RL line, attempts to add BL line - rejected (RL/BL strict separation). Saves DN-WH-A-20260617-001 for RL only.
Day 2 - 5:35 PM	Warehouse user	Repeats: new DN with only the BL line -> DN-WH-A-20260617-002.
Day 2 - 5:40 PM	Warehouse user	Outbound DNs -> submits both drafts to Logistics.
Day 2 - 5:45 PM	Logistics	Sees DN-WH-A-20260617-001 + 002 in their DN approval queue. Confirms delivery date Day 3 AM. Approves both.
Day 2 - 5:45 PM	(auto)	Both DNs move to awaiting-HOD-approval. The site HOD gets the "DN approved by Logistics" notification.
Day 2 - 6:00 PM	Site HOD	Logs in, sees red bell badge 2 unread. Opens inbox -> drills into the DNs via DN Approvals tab. Approves both.

When	Who	What
Day 2 - 6:00 PM	(auto)	Both DNs move to awaiting-Store-Keeper, and a matching draft receipt appears in the site's Receipt Staging for each. The site SK gets the incoming-DN notification.
Day 3 - 8:30 AM	Truck arrives at site GI-PS01	
Day 3 - 8:45 AM	Site SK	Opens Entry Log -> Receipt Staging. The new Incoming Delivery Notes from Warehouse expander is open at the top, showing both DNs. Inspects both, confirms physical match, clicks [OK] Mark as Received on each.
Day 3 - 8:45 AM	(auto)	Two receipts are posted to the ledger, each carrying its DN number, source warehouse and originating PO. Both DNs are marked received, the staged drafts are cleared, and the Live Dashboard shows the new stock immediately.

16.3 Side-path examples

A site rejects a DN

Same scenario, but on Day 2 at 6:00 PM, Site HOD inspects the line preview on DN-002 and notices the BL qty doesn't match the PR. Rejects DN-002 with note "BL qty in DN is 25, PR asked for 20 - bounce back".

- DN-002 is marked rejected, and the reason is stored with it
- Warehouse user gets the rejection notification with the reason
- The BL line on the PO frees up qty (the over-ship guard recalculates available since DN-002 is no longer "live")
- Warehouse prepares DN-003 with correct qty

Logistics force-closes the PR mid-flight

Project cancelled by management on Day 1 evening:

- Logistics -> Force-Close -> radio = PR -> reason "Project cancelled per CFO email 2026-06-16" -> Force-close PR
- Admin + Site HOD see critical-severity notification in their bell
- The pending PO can either stay (if vendor already shipped) or also be force-closed

- Site HOD's In-Transit -> Force-closures sub-tab shows the closure with the reason

Warehouse asks for reschedule

Receiving day Truck doesn't show up - vendor delay:

- Warehouse -> Outbound DNs -> drill into DN-001 -> Request reschedule -> new date Day 4 -> reason "Vendor delivery delayed 24h confirmed by vendor email"
- Logistics -> Reschedules -> reviews -> [OK] Approve with notes "Confirmed with vendor"
- PO Expected_Delivery, po_assignments.Expected_Delivery, AND DN_Date all flip to Day 4 in one transaction
- Site HOD sees the decision notification

16.4 What stops a duplicate PR or a double order

Added 2026-08 (Phase 8). Three rules run underneath the procurement chain. None of them changes how the screens are used; they change what happens when two people, or one impatient person, do the same thing twice.

16.4.1 A PR number is reserved, not guessed

Every PR number is claimed in a registry before the requisition's lines are written, so two HODs pressing Create PR in the same second get different numbers. If a number is somehow taken, the system takes the next one rather than failing. Renaming a PR moves its reservation with it and is refused if the new name is already in use - by a registered PR or by one that was imported without ever being registered.

16.4.2 Each step happens once, in order

A requisition moves site_draft -> submitted -> in_po, and each move is checked against the state the row is actually in - not the state the screen was showing when it loaded. Consequences you will see:

- Submitting twice does nothing the second time. It used to accept lines that were already submitted, rewrite their timestamp and fire a second notification at Logistics.
- A PO cannot be raised twice over the same lines. The second attempt is refused with a message naming the lines, rather than silently succeeding.
- A PR may still carry several POs. Partial fulfilment splits one requisition across vendors or deliveries, so the lock is per line, not per PR.

16.4.3 A retry is not a second order

Submit PR, Create PO, Assign PO and Rename PR each carry a key minted when the form opens. Press the button twice, lose the connection and retry, or double-click - the system recognises the repeat and replays the first answer instead of doing the work again. Sending the same key with different details is refused, because that is a client bug rather than a retry.

[!] The buttons disappear rather than grey out. Once a PR is submitted, the Submit button is gone - not disabled. A greyed-out button invites a second click and a support call about why it "isn't working"; an

absent one says the step is done.

21. 2026-08 Feature Update - What Changed

Everything in this chapter is live. Grouped by what you would notice.

21.1 Reports and exports look the same everywhere now

Every Excel export - not just the Material Estimator's - now carries the branded layout: the GI logo, a "Report Generated / Generated By" band, a title bar and a framed, filterable table with the header frozen. Previously only three SME workbooks looked like that and everything else came out as a bare grid.

> If you have automation reading these files: the header row is now row 6 > and data begins on row 7 (rows 1-4 are the logo and meta band, row 5 the > title bar). Anything that assumed a header on row 1 needs adjusting.

PDF reports no longer overlap or cut off text. Columns are now measured against their actual content and share the page fairly, and long values wrap onto more lines instead of being truncated. Previously every column got an equal slice of the page regardless of need, so a long material description was drawn on top of the next column while Date and UOM wasted most of theirs - and anything past 24 characters was silently deleted.

QR badge and bin labels were fixed the same way: a long employee name used to run past the edge of its sticker onto the one beside it.

21.2 The Material Estimator understands your order data correctly

Ordered_Qty is the TOTAL quantity procured for the project, and Available_Qty is the part of it that has already arrived - available is a subset of ordered, not a separate pile beside it.

The estimator used to add the two together, which double-counted every unit already on the shelf. On real data that understated the buy list by 22,951 units across 22 of 30 materials, and on GI-8005763 - where all 143,000 ordered units had arrived - it read 286,000 and reported nothing to buy against a demand of 152,685, hiding a 9,685-unit shortage completely.

What you will see in the UI:

- Second-tier quantities now read "Pending Delivery" - the part of the order still on its way - never just "Ordered".
- Second-tier coverage reads "When delivered".
- Reports carry a Total Procured column beside them.

Feasibility is unchanged and still judges physical stock only: a tank cannot be built with a purchase order.

21.3 A view-only Auditor role

See §20.

21.4 You are signed out after 30 minutes of inactivity

Leave the app untouched for 30 minutes and it signs you out and returns you to the login screen. A warning appears two minutes before with a button to stay signed in, and any real activity - a click, a keystroke, a scroll - resets the clock silently.

This matters most on shared site terminals, where an unattended session was previously good until the browser was closed.

21.5 Search anything with CmdK

Press CmdK (Mac) or Ctrl-K (Windows) anywhere in the app.

The palette already jumped to pages; it now also searches your inventory live. Type a SAP code, part of a material name or a material code and the matching items appear under a Materials heading - pick one and you land on its Material Intelligence card. Results are scoped to what your role may see.

21.6 The Hub Assistant is faster and more accurate

The assistant used to be handed its entire allowed portion of this manual on every single question. For an Admin that was the whole ~180 KB document, which is slow to process; for everyone else it was the first 800 characters of each chapter, which is the wrong 800 whenever the answer is further down.

It now finds the passages that actually match your question and reads only those - a 97.7 % smaller prompt for Admins, and noticeably better answers for everyone, because the relevant paragraph is now present in full instead of being cut off.

A parsing bug was fixed at the same time: numbered comment lines inside the worked examples in the Operations chapter were being mistaken for chapter headings, which silently replaced chapters 1-4 - Introduction, Roles & Permissions, Login, and the Store Keeper Manual - with fragments of a restore procedure. Ask "how do I log in?" today and you get §3.1, the login screen.

The role filter is unchanged and still applied before retrieval: a Store Keeper's context physically cannot contain an Admin chapter.

21.7 Serialised tool and equipment tracking

Two hammers share one item code, so scanning either label used to resolve to the same inventory line - the system could not tell you which one was in your hand or where the other had gone. Assets solves that.

Each physical item gets its own record, identified by its serial number. A scan now resolves three ways, and the order is deliberate:

What the label carries	What you get
A serial number or asset tag	That exact item, immediately - there is nothing to choose

What the label carries	What you get
Only the item code, and several units exist	A "which one is in your hand?" list to pick from
Only the item code, and no units are registered	The ordinary material view - the system does not invent an asset

Recording where something is. Open the item, tap Move, and either pick a rack or describe the place in words - not everything lives on a shelf, and "loaded on truck 4771" is a real answer. If the device can supply coordinates, they are captured alongside the update, never in front of it: a declined permission or a yard with no signal still records the move, with the coordinates simply absent. A recorded position becomes a map link on the item.

Status covers both what condition a tool is in - working, not in use, under repair - and where it sits in the stores workflow. The full history of every move is kept and is never overwritten, so "where has this been" is a question with an answer.

> Coordinate capture needs a secure connection. Over plain HTTP the browser > refuses to supply a position - the move still saves, without coordinates. If > your team is on the installed app or the company web address, this is already > the case.

> Your spreadsheet seeds this; the app owns it. A location typed in the > Consumption Log creates the asset on the next sync. From then on, what somebody > recorded in the app wins: a re-sync will not move a tool, revert its status, or > erase a captured position.

21.8 The warehouse rack locator

"Which rack is this in?" now has an answer. Search by material name, by code, or by scanning a label, and the system tells you the shelf to walk to - ordered so that the first result is where to go first, even when a material is stocked in three places.

- A material you stock but nobody has located comes back marked not located, rather than being left out. Silence would read as "we do not stock it", which sends somebody out to buy what is already on a shelf.
- Scanning the rack itself answers the reverse question - everything that is meant to be on that shelf. That turns a stock count from a hunt into a checklist.
- The locator is built into the CmdK search, so it is two keystrokes from anywhere in the application.

21.9 The Material Estimator handles messy tank references

The Consumption Log records a tank reference typed by hand, and the same vessel appears written four different ways. The system now recognises those as one vessel automatically - that alone accounted for 58 of 103 rows.

Where a reference is genuinely ambiguous, it refuses to guess. TNK-091 exists on both TRAIN J and TRAIN K; either answer would look completely plausible in every report afterwards, so those rows wait in a queue for someone who knows. Resolve the reference once on the Actual Consumption tab and every row waiting on it is tagged.

21.10 For administrators - local tooling

Two maintenance scripts ship with the system, documented in the administrator runbook: one puts the database and the public connection to sleep and wakes them again for a laptop that is not always on, and one takes and restores database snapshots, with an option to run nightly at 02:00.

> The previous nightly backup had been failing silently for 25 consecutive > nights - it pointed at a script that a repository reorganisation had moved. > Anyone relying on it had no local backups at all. The replacement is verified > by actually restoring into a scratch database on every change, which is the > only check that means anything.

21.11 Downloaded files now protect you from your own spreadsheet

If you type something that begins with an equals sign, a plus, a minus or an at-sign into a text field - a remark, a note, a reason - you may notice a small apostrophe in front of it when you open the downloaded Excel or CSV file.

That is deliberate, and nothing has been lost.

Spreadsheet programs treat a value beginning with one of those symbols as a calculation to run, not as words to show. That is fine when you write the formula yourself. It is not fine when the text arrived from somewhere else: a note typed at one site, exported by a manager and opened on their computer, would be run by their spreadsheet without anyone choosing to. The apostrophe tells the spreadsheet to show the text exactly as it was written instead.

What this means in practice:

- Numbers are never touched. Quantities, prices, areas and every total in every report stay real numbers, and every total still adds up.
- A plain number that happens to be negative, such as -5, is left exactly as it is - it is a number, not a calculation.
- Ordinary text - material names, supplier names, ordinary notes - is untouched.
- If you want the plain text back in your own copy, delete the apostrophe in that cell.

You are most likely to see this on remarks and reason fields, because those are the places people type freely. It applies to every download the system produces: reports, master-data exports and the Material Estimator workbooks alike.

21.12 Phase 9 - what changed for you

Five things changed at once, and one of them changes how you work every day.

21.12.1 [!] Consumption is filed from paper now

This replaces the store-keeper-first flow entirely. Full detail in §4.10.

You fill a printed form -> Store Keeper verifies -> HOD approves

- Print a form per lining system (§4.9). It lists every material for that system, so nobody writes a material name by hand - that is the one thing the reader is genuinely bad at.
- Fill in the date, equipment and area at the top, and a quantity and a lot number on every row.
- Photograph the whole page including the QR code and upload it.
- Check every figure. Where the handwriting was not certain the box is left empty rather than guessed, and those rows are marked.
- The store keeper verifies your quantities next, not the HOD. Their corrections show to the HOD in red, and you are told before it is approved.

[!] Approval now deducts the material as well as posting the area. Store keepers: stop raising a separate material issue for lining work - the execution entry is the only writer now, and doing both deducts the same drum twice. (From Phase 14 a drum that is also logged in the Excel Consumption Log is reconciled rather than deducted twice - see §6.17 - but the rule stands: one record per drum.)

[!] Rejection is final. Raise a new entry from a fresh form.

[!] Each printed sheet is used once, and a sheet printed before its system's materials changed will be refused. Both rules exist because your handwriting is matched to materials by row position.

21.12.2 WBS numbers, at last

The WBS # column was blank on every entry in the system - not because the feature was missing, but because there was no screen to reach it. There is one now: HOD -> WBS & Work Types (§16.5).

- Add your WBS numbers, then map each work type to one.
- An issue that leaves the WBS box blank gets the number its work type maps to.
- Work types are now a fixed list you curate, which is what stops civil and Civil becoming two cost centres.
- Nothing is enforced until you add the first row. Turning it on is your act.
- WBS is applied forward only; historical entries are not restamped.

21.12.3 A new chart: Efficiency by Day

Man-Hours -> Efficiency by Day (§19.6). For one lining system, how much manpower went into Equipment A against Equipment B - normalised on man-hours per m², which is the only figure that survives a change of size.

[!] The line is the running figure, not the day's, and it breaks on a day with hours and no area. Both are deliberate; §19.6 explains why.

21.12.4 The Manpower Planner tells you what nights actually buy

Three changes to the arithmetic:

- Nights buy calendar time, not a smaller payroll. The total headcount is unchanged - nobody works both shifts - but the plan now shows how many days the night crew saves.
- The day/night split follows your roster, not a division by two. This site runs about 20 on days against 80 on nights; halving the requirement understated the night crew fourfold.
- Hiring idle trades no longer flatters the plan: capacity counts only the roles the job needs.

21.12.5 "Labor" is now "Manpower"

The page, the menu entry and every heading now say Manpower - the word everyone here actually uses.

> If you have automation reading the API: the field names did not > change. Done_SQM_Labor and Labor_Variance_Pct are still spelled exactly > as they were. Only what you see on screen moved, so nothing you have built > against those keys needs touching.

21.13 Photo reading - the two failures that are fixed

If you tried to read a photographed Daily Consumption log or the new printed consumption form and it did not work, this is what was happening and what has changed. Nothing about how you use it is different - you still photograph the page and wait.

21.13.1 "The log came back empty, or said it could not be read"

The reader was being cut off part-way through a long sheet. It read your page correctly, got as far as about row 13 or 14 of a 30-row log, and ran out of room - and because the answer was incomplete, the system threw away all of it, including the thirteen rows it had read perfectly.

This is also why Delivery Notes always worked: a note is four or five items and always finished comfortably. It was never that the reader could not handle your handwriting or the free-form table.

Now: it is given enough room for a full 30-row sheet, and if a page is ever still too long, the rows it did read are kept and only the unfinished one is dropped. On the operator's own test photo this went from nothing at all to all 30 rows.

21.13.2 "The new PDF just hangs, then fails"

A full page of handwriting genuinely takes the reader several minutes. The system was giving up after four - often within seconds of it finishing - and then telling you "the vision model is not reachable", which sent people off to check a service that was running perfectly.

Now: it waits long enough for a real page, and if something genuinely is wrong it says which of the two it is: still working, or actually unreachable.

21.13.3 Two smaller things you may notice

- A quantity the reader could not make out now arrives BLANK, not as 0. It used to show a confident zero, which reads like a number somebody wrote and nobody re-checks. A blank box is a question, and the row will not submit until you answer it.
- A photo that produced nothing now says so. Previously a failed read could finish looking merely "empty", which is indistinguishable from a photo of a blank form. If the page could not be read, you are told to retake it.

21.13.4 If a photo still will not read

Nothing has changed about the advice, but it is worth repeating: get the whole page in frame, keep the lighting even, and hold the camera square to the paper. The Paste tab works with the AI switched off

entirely and is always available. Typing the entry in by hand is never wrong.

21.13.5 [OK] Ditto marks now come through (2026-09-02)

Where: Entry -> OCR Import (/entry/ocr), the Consumption log lane.

If you have been retyping the Name, Tank No. and Product Name columns after every photo read, that was a real fault and it is fixed.

Most people write a ditto mark - " or (ditto) - instead of repeating themselves down a column, and on a typical sheet that is most of the page. The reader was seeing those marks perfectly well and quietly returning nothing for them, because to a reader "same as above" is not a word to transcribe. The step that was supposed to copy the value down was looking for the mark, found an empty cell instead, and left it empty. On the sheet this was diagnosed against, that was 19 of 30 tank numbers, 14 of 30 names and 8 of 30 product names.

Now: a dittoed cell is filled in from the row above, whether the reader returns the mark or returns nothing.

Two things worth knowing about how it behaves:

- A cell that was filled in for you is marked [?]. If the reader gave us nothing and we worked the value out from the row above, the row carries an information marker saying so. A cell where you actually wrote a ditto mark is not marked, because that one is what the paper says rather than what we concluded. It never blocks anything - it just lets you see the difference.
- Rows you never wrote on stay empty. The S.No. column is printed on all thirty rows whether or not anybody used them, so if you filled in ten rows, rows 11 to 30 come back blank. They are not filled in with the last person's name.

Nothing about how you use the page changed: photograph it, press Validate against the spec, review, stage.

22. Quality, Safety, Employees & Procurement (QSEP)

Everything in this chapter went live in August 2026. It adds four things that used to live on paper or in somebody's memory: who checked the material, who is wearing the safety gear, where a person works, and what the scanned purchase paperwork actually said.

Read section 22.1 first even if only one of the four concerns you - the QC approval rule changes what happens when a Store Keeper tries to issue certain materials, and that affects the whole site.

22.1 Quality Control - the new role and the issue block

What a QC does

A Quality Control inspector checks material that carries risk if it is wrong. Today that means the Surface Shields category - 36 materials out of the 466 in the master list. Nothing else in the catalogue is affected by any rule in this section.

> ### The rule, in one line > > Material without an MTC CAN be sent and dispatched to site. It CANNOT be > issued or consumed at the site. > > Receiving it is fine. Putting it on a Delivery Note and driving it to site is > fine. Handing it to a worker is not. If you have been told otherwise, that > came from a document written before 2026-08-12 - the rule below is what the > system actually does.

There are two separate gates. Both apply at the same moment - the instant before material goes to a worker - but they are satisfied by different people, which is what makes them worth telling apart:

Gate	When it applies	What it demands	Who is stopped	Who fixes it
Material Test Certificate	When a Store Keeper issues material to the field	A certificate on file covering that material at that site	The Store Keeper at the issue form	Logistics (or the warehouse, or the SK)
QC approval	The same moment	A QC must have inspected and approved that quantity at that site	The Store Keeper at the issue form	The QC

Material may be received, and may travel to site, with neither. That is deliberate and was decided by the operator. Holding a truck at the gate for paperwork that is sitting in somebody's inbox costs more than it saves - and worse, refusing to record a delivery does not stop it arriving; it only stops the system knowing about it, so real material sits in a yard invisible to the stock report and to everyone planning around it. What material may not do is reach a worker's hands unchecked. So both gates bind at issue.

When a controlled item is received without a certificate, Logistics is notified automatically to chase it from the supplier. Nobody has to remember.

You should almost never upload the same certificate twice

The certificate follows the material down the chain. Whoever has it files it once:

- Logistics attaches it to the purchase order line - before the goods have even shipped, and without knowing which site will end up with them;
- the warehouse attaches it to the Delivery Note as the material leaves;
- the Store Keeper uploads it at the site, from the Receive form or from the banner on the Issue form.

Any one of those clears the material for issue at the destination site, and the issue form tells you which document cleared it and where it came from. A certificate filed for one site does not clear another - it attests to one batch from one production run.

If you find yourself uploading a second copy of a document the company already has, stop and report it. That is a fault, not a workflow.

Raising an inspection

An inspection is opened automatically when controlled material is received. The QC does not create it by hand. It appears in Quality -> Inspections with a status of pending.

Deciding an inspection

Open the inspection and enter the quantity you approve. The status follows from the number, not from a separate choice:

You approve	Status becomes
Nothing (zero)	Rejected
Less than the submitted quantity	Partially approved
The whole submitted quantity	Approved

Rejected material stays in stock. It is not automatically sent back to the vendor and it does not disappear. It sits where it is, unusable, and is blocked from issue until somebody decides what happens to it. That was an explicit instruction: an automatic vendor return removes the evidence before anyone has looked at it.

What the inspector sees, and the Return No (added August 2026)

The inspection screen now shows the material's name, with the SAP and material codes underneath it. Before this it showed only the codes, which meant judging a material by a number rather than by what it is.

If a Material Test Certificate is on file, the certificate number appears with an Open certificate link that opens the actual document - in the queue and again in the inspection dialog. Previously it only said that a certificate existed, with no way to read it, which meant approvals were being made against documents nobody had opened.

When an inspector rejects any quantity, the system issues a Return No such as QCR-20260813-41. It is announced to the inspector on the spot, listed against the inspection, and sent to both the site's Store Keeper and its Head of Department. The Store Keeper types it into Return Stock and the return form fills itself in - see §13.10.

> This does not change the rule above. The Return No is an invitation for a > person to raise a return; nothing moves, and nothing is sent back to a vendor, > until the Store Keeper posts it and the HOD approves it.

The daily missing-certificate alert (added August 2026)

Every morning the system checks for Surface Shield material that is in stock with no Material Test Certificate on file, and tells the people who can do something about it:

Where the material is	Who is told
In a warehouse	Logistics, the Warehouse User, and the warehouse's QC
At a site	That site's Store Keeper, HOD and QC, and Logistics

Logistics appears on both lists because they are the only people who can obtain the document from the supplier. One message per location lists everything outstanding there, rather than one message per material.

The alert repeats every morning until the certificate is uploaded, and then stops on its own. That repetition is intended: this is a standing condition, not a one-off event, and the material cannot be issued to anybody while it lasts.

What the Store Keeper sees when the block fires

The issue form refuses with a message that names the actual numbers, so nobody has to guess which of three situations they are in:

- No inspection at all - "no quality inspection exists for it at this site. It cannot be issued to the field until a QC has checked the material and its certificate."
- Inspected but nothing approved yet - names how many inspections are still pending.
- Partly approved and used up - "QC has approved 40 and 40 is already issued or staged, leaving 0 - not enough for 10. Ask the QC to inspect the remaining stock."

The clearance is pooled per material per site, not per individual pallet. What is counted against it is everything already issued or staged awaiting HOD approval, from the date of the first inspection onward - issues that predate quality control entirely are not counted, or the site would be frozen forever by its own history.

> This does not change the FEFO or over-issue behaviour. Those remain > warn-and-record, as they always have. The quality block is about whether the > material is fit to use, which is a different question from whether the shelf > count agrees.

Creating a QC account

Quality -> QC Accounts, available to an HOD, a Warehouse User or Logistics. You create a QC inside your own scope: an HOD creates one for their site, a Warehouse User for their warehouse. Set a site or a warehouse, never both.

Moving a QC to another site is a request, not an action - the HOD raises it and an Admin decides. A QC whose remit could be changed by the person whose work they inspect is not an independent check.

22.2 PPE - issued through the ordinary form

There is no separate "Issue PPE" page, and this is deliberate. Safety boots leave the store the same way a drum of resin does: through Entry Log -> Issue. The form recognises that the item is PPE and asks for the extra details on the spot.

Select a PPE item and three fields appear:

Field	When it is required	Why
Employee ID number	Always	PPE is tracked against a person, not a site
Safety Approval document	Unless a usable-time rule says otherwise	Somebody signed for this issue
Reason for early replacement	Only if the worker's current item has not expired yet	A boot replaced after three weeks is either a defect or a leak

The stock movement and the PPE record are written together. There is no second ledger and no separate PPE stock - the quantity leaves through the normal issue path, so stock levels, burn rate, reports and every other number stay correct without knowing anything about PPE.

The messages you will see

- "...is PPE - name the employee receiving it (their ID number), so it can be tracked against them."
- "worker 30816 is not in the employee master" - the person must exist on the roster first.
- "worker 30816 is at site RIYADH, not CNCEC - transfer them first if they have moved" - see 22.3.
- "Ahmed already holds this item, issued 2026-07-02 and good until 2026-10-02 - give a reason for replacing it early."

Usable time

Safety & People -> PPE Usable Time (Store Keeper or HOD) is where you say how long an item lasts: 90 days for gloves, 365 for a helmet, and so on. A rule can be global or set for one site, and the site's rule wins where both exist.

> [!] Until a rule exists for an item, two things follow. The issue will > always demand a Safety Approval document, and the item gets no expiry date > - which means it will never appear in the forecast below. If the PPE Forecast > looks empty, this is almost always why.

An expiry is a suggested replacement date, not a restriction. Nothing stops a worker using gear past it, no alert is sent to anybody's phone, and the worker is not blocked from anything.

The 15-day forecast

Safety & People -> PPE Forecast lists PPE expiring in the next fifteen days and tells you what to order:

> suggested = expiring - what is on the shelf - what is already on order

If the answer is zero because thirty units are already on an open purchase order, that is the correct answer and not a broken screen. The list shows the names of the people whose gear is expiring,

because a column of quantities cannot be sanity-checked by a human and a column of names can. A 90-day issue rate is shown beside the suggestion as a cross-check, never folded into it.

22.3 Employees - one person, one number

The employee ID number is the person. It is unique across the whole company, not per site, and everything else hangs off it.

Safety & People -> Employees shows the roster. An HOD can transfer someone to another site, and the transfer takes effect immediately - no approval step, because a person who has physically moved has already moved.

PPE history follows the person, not the site. Transfer a worker from one site to another and the boots they are wearing are still on their record. This falls out of keying on the ID number rather than being a feature bolted on top, which is why it cannot quietly stop working.

Admins get a timeline for any employee: every site they have worked at, with dates, and everything they currently hold. Use it when somebody asks "who had this, and when".

Employees -> Data Quality lists records that cannot be used and says why - a missing ID number, a duplicate, an inactive worker still holding gear. Check it before a PPE rollout rather than discovering the gaps one refused issue at a time.

22.4 Procurement - drafted delivery notes, urgency, and scanned paperwork

Delivery Notes draft themselves

When a warehouse receives goods against a purchase order, the system now prepares the Delivery Notes the sites are waiting for, grouped so that R/L and B/L material never share a note.

They are created as drafts. A person still adds the vehicle and driver and submits them, because the system cannot know that a truck exists. A notification tells the warehouse the drafts are waiting.

If a draft cannot be created - an over-shipment, a missing certificate - the goods receipt still stands. The stock genuinely arrived; losing that record because a convenience failed would be far worse. The reason is written to the audit log and the clerk cuts the note by hand.

Auto-drafting can be switched off in Settings.

Assigning a purchase order to a warehouse (changed August 2026)

A purchase order goes to one warehouse. The Purchase Orders list now shows a Warehouse column naming where each order was sent, and once an order has been assigned the Assign button is replaced by an assigned tag.

Previously the button stayed on every row for ever, and the list gave no indication that an order had already been routed. Pressing it a second time created a second assignment and told a second warehouse to expect the same goods, with nothing anywhere reporting a problem.

- Assigning the same warehouse again - a double-click, or an old tab - is accepted and changes nothing. No duplicate, no second notification.
- Assigning a different warehouse is refused, and the message names the warehouse that already holds the order. Re-routing an order another warehouse is already expecting is a decision for a person to make, not something the system will do quietly.

Shipping a delivery note (changed August 2026)

Pressing Ship now asks for two things before the delivery leaves:

- The delivery note number printed on the physical document - the carrier's or the supplier's number. This is not the DN number the system generated; the point is to connect the two, so that a delivery can be reconciled later without a telephone call.
- A scan or a photograph of the signed note. The camera button opens the phone camera directly, which is the practical option at a loading bay.

Both are required. If either is missing you are told which one, separately - one is typing and the other is scanning, and being sent to redo the half you had already done is its own small waste.

A Material Test Certificate is still optional here and works exactly as before. A certificate covers the material and is inherited from the purchase order or the delivery note down to the receiving site, so nobody uploads it twice. A delivery note covers this shipment and is inherited by nothing. They are different documents answering different questions.

Once shipped, the document number and a download link appear beside the delivery for everyone who touches it - the warehouse, Logistics, the HOD, Quality and the receiving Store Keeper. The Store Keeper is the one holding the paper copy while the truck waits, so this is where it matters most.

> Deliveries shipped before this change show "not shipped yet" in that > column. There is no document for them because none was ever captured, and > inventing a number would be worse than saying so.

Marking a reschedule urgent

A delivery reschedule can be marked urgent. An urgent one goes out immediately instead of waiting for the 4 p.m. evening digest. Use it sparingly - a channel where everything is urgent is a channel nobody reads.

Reading a scanned PR or PO

Upload the supplier's paperwork and the system reads the lines off it, so nobody retypes forty rows.

The file is stored before it is read, and this matters more than it sounds: the document that defeats the reader is precisely the one somebody will need to look at later. A failed read still leaves you the document.

There are two ways a document gets read, and the system chooses by whether the file actually contains text - not by whether it is a PDF:

The file	What happens
A PDF exported from a system	Text is read directly. Fast, exact.

The file	What happens
A PDF that is a photograph or a scan of a signed page	Read by the vision model. Takes longer; you poll for the result.

A printed-signed-and-scanned purchase order contains no text at all, only pictures of text. Before this change such a file was reported as read successfully with zero lines found - which is worse than an error, because nothing on screen distinguished "this order has no items" from "I could not read a word of it".

Once read, confirming the document links it to the purchase request or order you created, so the original scan is one click from the record. Offering a purchase order scan to a purchase request is refused by name - the two are not interchangeable.

> A quantity the reader could not make out is left blank, not zero. A zero on > a purchase order looks like an answer and would be ordered as one.

22.5 Assets - one hammer, one row, anywhere in the company

A serialised asset is now unique by its part number and serial number across the entire company, not per site. The same hammer cannot be registered twice by two sites who both think they have it.

Moving an asset between sites is a request approved by the HOD of the site it is leaving - the site giving something up is the one that needs to agree. On approval the asset's site changes and its old rack assignment is cleared, since a shelf in one yard means nothing in another. The move is recorded as a movement, so "where has this been" still answers correctly for the leg between the two sites.

Trying to re-register a serial that exists elsewhere is refused with a message that says where it actually is and what to do about it.

22.6 Passwords

The minimum password length is 8 characters, and a password must now also contain an uppercase letter, a number and a special character. This replaced a flat 12-character minimum with no composition requirement.

The rule is applied identically everywhere a password is set - admin creation, password reset and self-registration. Self-registration previously enforced a much weaker rule than the admin screens did.

The Request Access page now offers Quality Control in its role list.

26. Practice Mode - learn the system without touching Live

Practice is a separate copy of GI Hub for training. It has every page, every role and every workflow Live has, but it runs on its own made-up data in its own database. Nothing you do in Practice can reach Live: not a receipt, not an approval, not a message.

Use it to learn a job before doing it for real, to rehearse something you do rarely (a return, a stock count, a PR), or to try something out without worrying about the result.

26.1 Switching between Live and Practice

On the sign-in screen, above the username box, there is a Live | Practice switch.

- Live is selected unless you choose otherwise. It is the real system.
- Choose Practice and the page reloads, ready to sign in to Practice.
- Choose Live again to go back.

Things to know about switching:

- Switching signs you out. You sign in again in the environment you picked. This is deliberate: a browser is in one environment at a time.
- Your Live password does not work in Practice, and a Practice password does not work in Live. They are separate systems with separate accounts (§26.3).
- The installed apps (Windows, macOS, Android, iOS) have the same switch.

26.2 How you know you are in Practice

The page tells you in six ways, and none of them can be dismissed:

Where	What you see
Colours	Practice is violet: the sidebar, the header, the page background and the main buttons. Live is navy and gold.
Top left, under the GI Hub logo	A pulsing amber PRACTICE badge, on every page. On a phone it sits beside the menu button. On the sign-in page it appears at the top left as soon as you choose Practice. With Reduce motion on, it stays still.
Top of every page	An amber bar: PRACTICE - practice data, nothing here is real.
Page header	An amber PRACTICE tag beside the title, and an amber line under the header that stays when you scroll
Browser tab	The tab title starts with PRACTICE -

Where	What you see
Anything you print	A large diagonal PRACTICE - NOT REAL DATA watermark

These come from the server you are connected to, not from the switch. The server reports which environment it is, and the page shows what it says.

[!] If you see a red bar saying "Environment mismatch", the switch and the server disagree. Everything you try to save will be refused until that is fixed. Sign out, pick the environment again, and tell your administrator if the red bar comes back.

26.3 The practice accounts

Practice has one shared account per role. Your trainer gives you the password for these accounts.

Role	Practice account
Store Keeper	practice.storekeeper
Supervisor	practice.supervisor
Head of Department	practice.hod
Quality Control	practice.qc
Head of Qualities	practice.qchod
Warehouse	practice.warehouse
Logistics	practice.logistics
Auditor (view-only)	practice.auditor
Admin	practice.admin (separate password, held by your trainer or administrator)

The site-based accounts work at site CNCEC, and the Warehouse account works at warehouse WH-01.

Everyone in a class uses the same accounts, so you will see each other's practice entries. That is normal.

26.4 What is in Practice

The data is made up but realistic. The categories, units of measure, lining systems, work types and tank numbers are real. The material names, SAP codes, people, vendors, quantities, prices and dates are all invented. Nothing in Practice is a real person or a real stock figure.

It is the same data the training videos use. If a video shows Tutorial Demo Gasket Set (SAP 899001), you can find it in Practice.

Work is already waiting so every role has something to do: receipts, issues and a return waiting for HOD approval, a draft purchase request, and one already with Logistics.

Every new Live feature comes with a Practice example (rule 17g), so it can be tried before it is used for

real. The current ones:

Feature	Try it on
The store keeper's note filling the job card (§4.9a.2)	One tank and day with two notes, Floor - 12.5 SQM Done and Sump Wall - 4.2 SQM Done: two jobs, each with its area and part filled in.
Lots, FEFO and expiry (§3.10.4)	PRACTICE PU PRIMER (899971): PR-OLD (expired), PR-SOON (the FEFO lot), PR-LATE.
CHEMOLINE rolls (§3.10.4)	PRACTICE CHEMOLINE (899973), batch 1026009999, three rolls.
Lots used but never received (§3.10.3)	Lot PR-TYPO.
The assistant's quick answers (§3.12)	As practice.hod, ask the Hub Assistant "which lots expire in the next 30 days?" - a table, from Practice's own lots. As practice.storekeeper, ask "open the lots page" - an Open Lots & Expiry button - and "is there a video on staging a return?". Nothing new had to be added: the quick check reads whatever database the Practice process holds (rule 17).

26.5 What works differently in Practice

Everything works exactly as in Live, except these four things. Each one is switched off on purpose:

In Practice...	Why
No WhatsApp message or email is ever sent. Notifications still appear in the bell.	A practice alert must never reach a real phone or inbox.
Photo reading (OCR) is switched off. The page explains this, and you can still paste text or type rows in.	The AI engine is shared with Live and takes minutes per page, so trainees must never queue ahead of real forms.
Training videos watched in Practice are not counted. You can watch as much as you like, but I have watched and understood this is refused.	Training records are kept in Live only. Sign in to Live to have a video counted.
Two-factor authentication is off, and cannot be turned on.	The accounts are shared, so one person's authenticator would lock everyone else out.

26.6 Offline entries stay where they were made

If you lose signal while saving an entry, GI Hub keeps it and sends it when the connection returns (§3.7). In Practice this still works, with one extra guarantee: an entry saved in Practice is only ever sent to Practice. If you switch to Live before it has been sent, it waits, and it is sent the next time this browser is back in Practice. It can never be sent to Live, even if you sign in to Live on the same device.

26.7 Resetting Practice (admins)

Over time Practice fills up with everyone's practice work. An admin can put it back to its starting state:

1. Sign in to Practice as practice.admin.
2. Open Admin Console -> Practice. The page shows when the data was built and when, and by whom, it was last reset.
3. Click Reset Practice data... and type RESET PRACTICE DATA to confirm.

The reset takes about a second. It removes everything anyone has done in Practice, and everyone signed in to Practice is signed out. The approval queues are filled up again. Live is not affected in any way. It is a different database, and the Practice server has no permission to open it.

The Practice tab and the reset only exist in Practice. In Live there is no such button, and the server has no such function.

How administrators set up and run Practice on a server is in §17.9.